

PUBLIC
EXHIBIT 55

U.S. Patent No. 10,707,167 Domestic Industry Chart

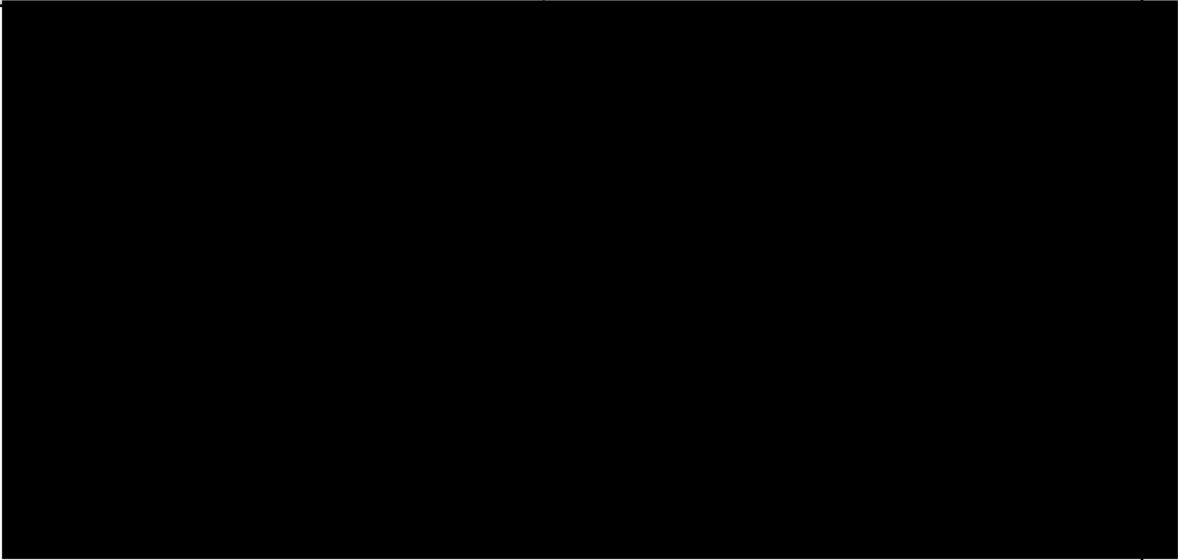
Title: Contacts to semiconductor substrate and methods of forming same

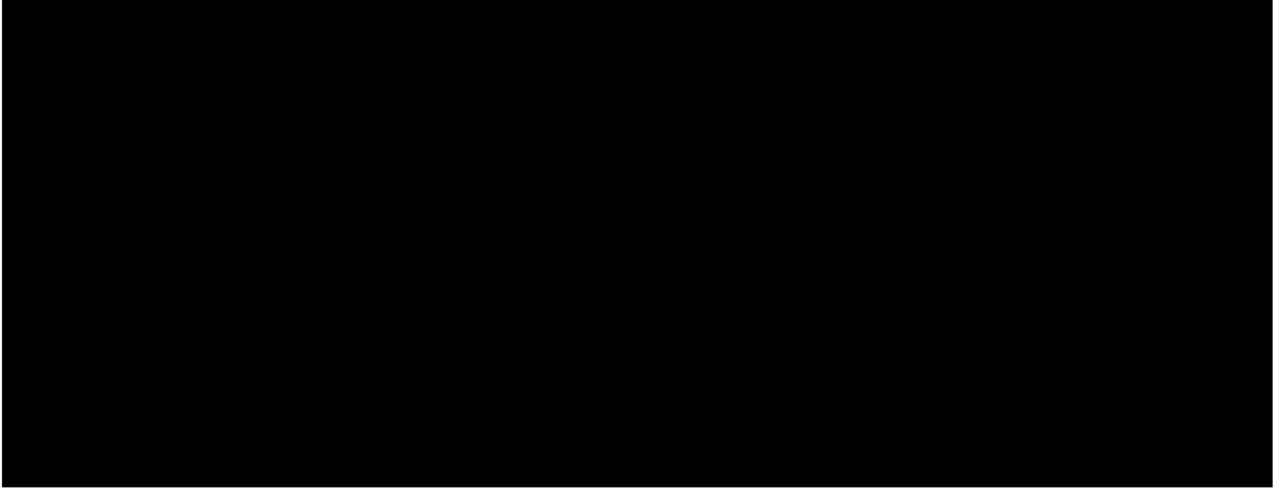
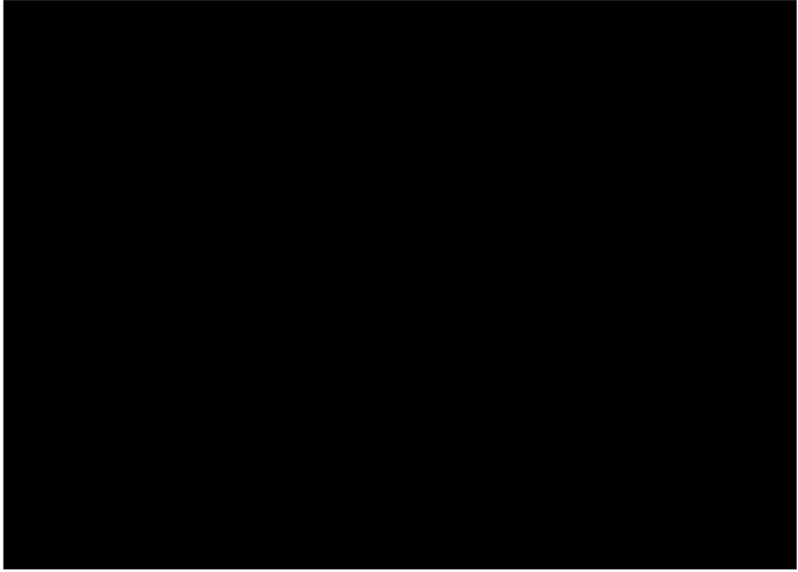
Inventors: Emre Alptekin, Nicolas L. Breil, Christian Lavoie, Ahmet S. Ozcan, Kathryn T. Schonenberg

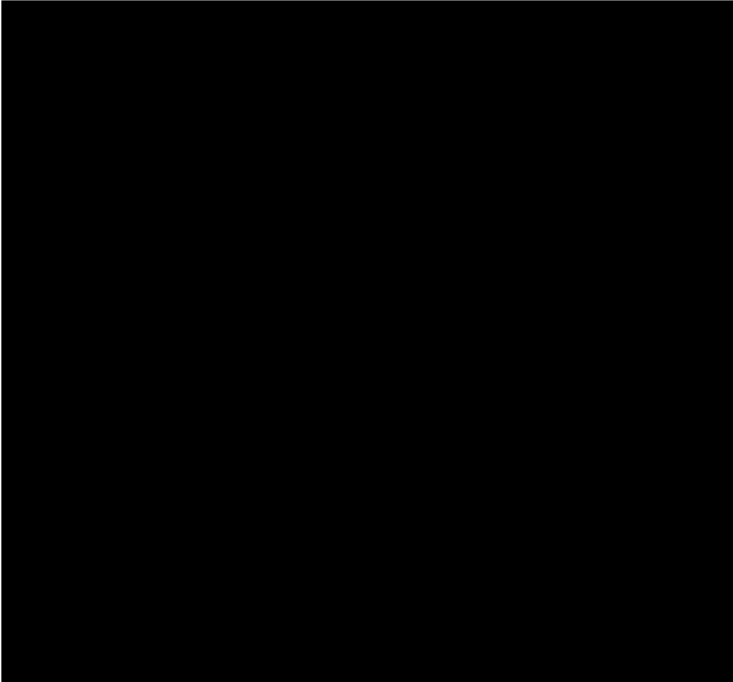
Abstract: An aspect of the invention includes a method for forming a contact in a dielectric layer over a semiconductor substrate. The method may comprise: forming a contact opening in a dielectric layer over the semiconductor substrate to expose an upper portion of the semiconductor substrate; depositing a first liner layer to conformally coat the contact opening; causing a portion of the first liner layer to diffuse into the upper portion of the semiconductor substrate to form a first intermix region at the upper portion of the semiconductor substrate; depositing a refractory metal layer over the first intermix region; and depositing a metal in the contact opening thereby forming the contact.

Claim	'167 Domestic Industry Products
[1pre] A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:	To the extent the preamble is limiting, chips manufactured using GlobalFoundries' 130CBIC Process (hereinafter the "'167 Domestic Industry Products") ¹ comprise a semiconductor structure including a contact to a semiconductor substrate.

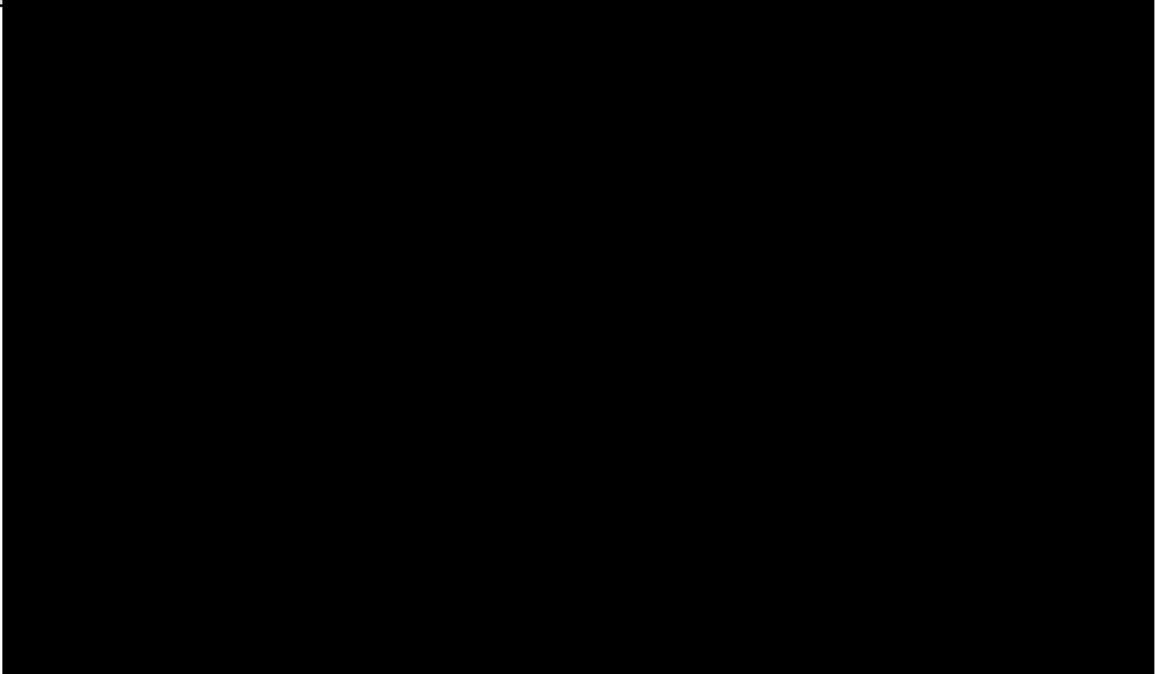
¹ GlobalFoundries reserves the right to identify different or additional '167 Domestic Industry Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances.

Claim	'167 Domestic Industry Products
	
[1a] a contact opening in a dielectric layer over the semiconductor substrate;	The '167 Domestic Industry Products comprise a contact opening in a dielectric layer over the semiconductor substrate. For example, the TEM images below show a contact opening after it has been filled (in black).

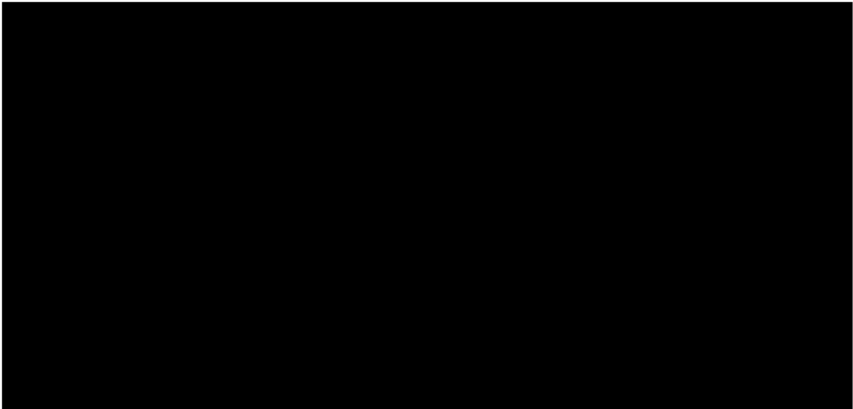
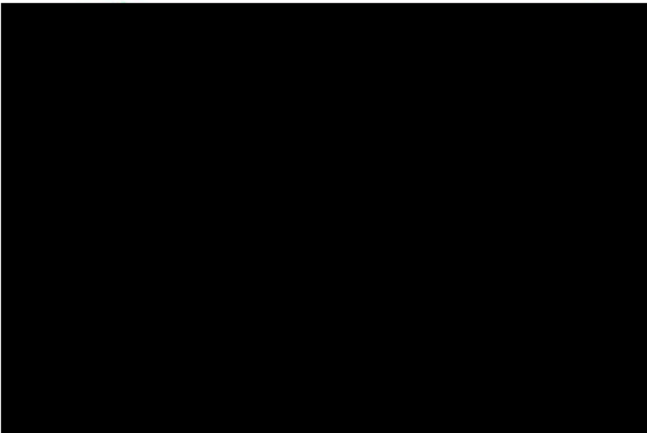
Claim	'167 Domestic Industry Products
	 As shown below, the contact opening is in a dielectric layer over a silicon substrate. 

Claim	'167 Domestic Industry Products
[1b] a first liner layer substantially coating a sidewall of the contact opening;	The '167 Domestic Industry Products comprise a first liner layer substantially coating a sidewall of the contact opening. For example, the first liner layer in the '167 Domestic Industry Products can be seen in the material analysis below.  The first liner layer substantially coating the sidewalls of the contact opening can be seen more clearly in the material analysis below, which separates the elements above.

Claim	'167 Domestic Industry Products
[1c] a refractory metal layer substantially coating the first liner layer in the contact opening;	The '167 Domestic Industry Products comprise a refractory metal layer substantially coating the first liner layer in the contact opening. For example, the first liner layer in the '167 Domestic Industry Products is substantially coated by a refractory metal layer, as shown in the material analysis below.
[1d] a metal substantially filling the contact opening; and	The '167 Domestic Industry Products comprise a metal substantially filling the contact opening. For example, the material analysis below shows a metal substantially filling the contact opening.

Claim	'167 Domestic Industry Products
	
[1e] an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate;	The '167 Domestic Industry Products comprise an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate. For example, the intermix region in the '167 Domestic Industry Products is shown in the material analysis below. The green arrow in the image immediately below corresponds to the green image at the top of the ensuing material analysis chart.

Claim	'167 Domestic Industry Products

Claim	'167 Domestic Industry Products
	For example, the images below shows the material stack for the source/drain region. The material analysis charts below show the composition of the source-side material stack. The green arrow pointing upwards in the figure below corresponds to the green arrow pointing from right to left in the following figure.  
[1f] wherein the refractory metal layer	In the '167 Domestic Industry Products, the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material

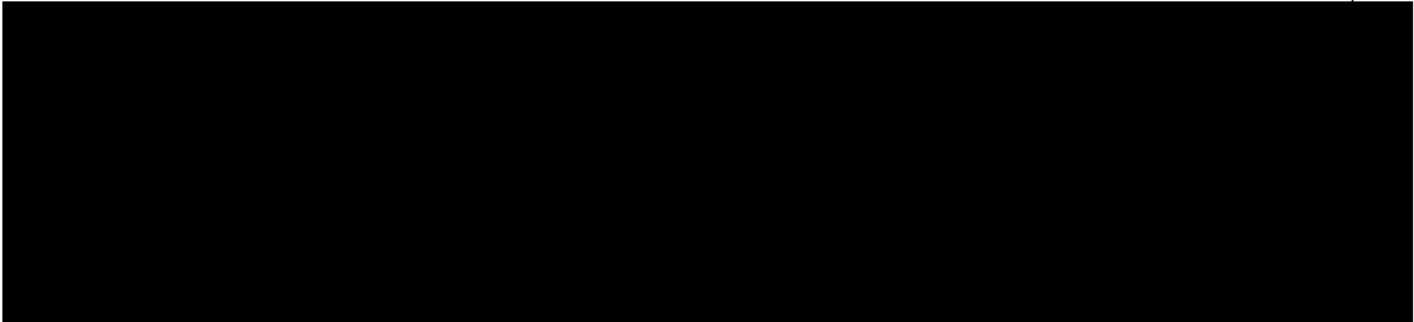
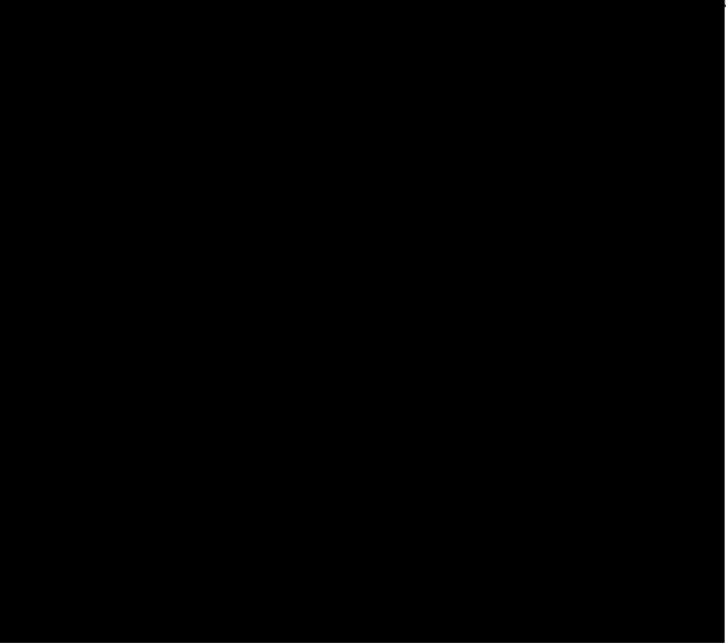
Claim	'167 Domestic Industry Products
includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.	substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal. For example, the refractory metal layer in the '167 Domestic Industry Products is shown in the images below and includes a first and second refractory metal material.  In the '167 Domestic Industry Products, the first refractory metal material separates the first liner layer and the second refractory metal material. The second refractory metal material, in turn, substantially separates the first refractory metal material and the metal contact.

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